

Qualification Pack



Software Programmer

Electives: Java/ Python/ HTML 5/ Kotlin/ Golang

QP Code: SSC/Q0510

Version: 2.0

NSQF Level: 4

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SSC/Q0510: Software Programmer

Brief Job Description

The Software Programmer qualification trains individuals in fundamental programming principles, software development methodologies, and a specific programming language (Java, Python, HTML5, Kotlin or Go). It also includes essential skills in project management, requirements engineering, and general employability/life skills, and offers specialization through elective modules.

Personal Attributes

They must show enthusiasm to learn new things, they must have good communication skills to communicate with colleagues and stake holders

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N0509: Verify the specifications and contribute to the design of software program](#)
2. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Electives (mandatory to select at least one):

Elective 1: Java

This elective deals with Programming in Java

1. [SSC/N0510: Develop, test and execute software programs as per specifications using Java](#)

Elective 2: Python

This electives deal with Programming in Python

1. [SSC/N0511: Develop, test and execute software programs as per specifications using Python](#)

Elective 3: HTML 5

This electives deals with Programming in HTML 5

1. [SSC/N0512: Write, test and execute programs in HTML5 to develop web or mobile applications as per specifications](#)

Elective 4: Kotlin

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This elective deals with Programming in Kotlin

1. [SSC/N0515: Develop, test and execute software programs as per specifications using Kotlin](#)

Elective 5: Golang

This elective deals with Programming in Golang

1. [SSC/N0516: Develop, test and execute software programs as per specifications using Golang](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	IT Services
Occupation	Application Deployment
Country	India
NSQF Level	4
Credits	40
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2512.0100
Minimum Educational Qualification & Experience	12th grade Pass (with computer background) OR 10th grade pass with 3 Years of experience in Programming languages (Such as Python, HTML, CSS, JavaScript, Golang etc.).The relevant experience would include work, internship, and apprenticeship after completing relevant educational qualifications. OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience in Programming languages (Such as Python, HTML, CSS, Javascript etc
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA

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Next Review Date	16/04/2029
NSQC Approval Date	16/04/2026
Version	2.0
Reference code on NQR	QG-04-IT-ITeS-05177-2026-V1-NNASSC
NQR Version	1

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SSC/N0509: Verify the specifications and contribute to the design of software program

Description

This NOS is about understanding the business and technical requirements of the software program to be developed

Scope

The scope covers the following :

- Requirement Analysis, Design Analysis, Best Practices

Elements and Performance Criteria

Requirement Analysis

To be competent, the user/individual on the job must be able to:

PC1. Understand the business context and business/ end-user requirements for the program

PC2. Understand functional and non-functional requirements for the program

Design Analysis

To be competent, the user/individual on the job must be able to:

PC3. Interpret the high-level and low-level design parameters using appropriate sources

PC4. Participate in the design of software program

Best practices

To be competent, the user/individual on the job must be able to:

PC5. Understand the project management frameworks used (agile methodologies and design thinking principles) in planning the software program

PC6. Keep a track of latest technological developments related to programming and identify their potential applications in your business domain

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational procedures and guidelines related to SDLC programming and documentation
- KU2.** Access and utilize data source documents and information relevant to software development
- KU3.** Identification of stakeholders involved at various stages of the SDLC.
- KU4.** Availability and effective use of standard templates and platforms for coding documentation and project management
- KU5.** An overview of Industry 4.0 technologies their use cases and potential applications
- KU6.** Basic concepts of design thinking agile methodologies and iterative development approaches

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- KU7.** Programming constructs coding practices and standards relevant to the chosen programming language
- KU8.** Tools platforms and frameworks necessary for code development testing and deployment in the programming environment
- KU9.** Fundamental techniques for securing software code against vulnerabilities
- KU10.** Tools and methods for documentation data visualization analysis and reporting
- KU11.** Principles of requirements analysis and translating business needs into technical specifications

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Understanding the impact of business related decisions on the organization
- GS2.** Communicating effectively both orally and in writing to convey information clearly
- GS3.** Planning and organizing work efficiently to achieve targets and meet deadlines

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Requirement Analysis</i>	10	22	-	8
PC1. Understand the business context and business/ end-user requirements for the program	5	11	-	4
PC2. Understand functional and non-functional requirements for the program	5	11	-	4
<i>Design Analysis</i>	10	22	-	6
PC3. Interpret the high-level and low-level design parameters using appropriate sources	5	11	-	3
PC4. Participate in the design of software program	5	11	-	3
<i>Best practices</i>	10	6	-	6
PC5. Understand the project management frameworks used (agile methodologies and design thinking principles) in planning the software program	5	3	-	3
PC6. Keep a track of latest technological developments related to programming and identify their potential applications in your business domain	5	3	-	3
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0509
NOS Name	Verify the specifications and contribute to the design of software program
Sector	IT-ITeS
Sub-Sector	IT Services
Occupation	Application Deployment
NSQF Level	4
Credits	3
Version	2.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

SSC/N0510: Develop, test and execute software programs as per specifications using Java

Description

This NOS is about program development and execution in java aligned to the prescribed technical specifications

Scope

The scope covers the following :

- Design Verification, Programming and Testing, Documentation

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Check the design specifications and define the objectives and functionality of the Java program and align them with design specifications
- PC2.** Develop software code for various applications in Java
- PC3.** Implement coding best practices such as reusability and naming conventions etc.
- PC4.** Create appropriate test cases and test the Java program
- PC5.** Debug the code and review the code with appropriate stakeholders
- PC6.** Maintain a record of test cases and exception handling
- PC7.** Leverage Gen AI tools to automatically generate code snippets for Java programs based on prompts and requirements.
- PC8.** Use Gen AI powered testing tools to create and validate comprehensive test cases for the Java application
- PC9.** Employ Gen AI driven debugging prompts and suggestions to identify bugs and troubleshoot issues efficiently
- PC10.** Utilize Gen AI prompts and suggestions to support automated code reviews and documentation processes

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational procedures and guidelines related to SDLC programming and documentation
- KU2.** Access and utilize data source documents and information relevant to software development
- KU3.** Identification of stakeholders involved at various stages of the SDLC.
- KU4.** Availability and effective use of standard templates and platforms for coding documentation and project management
- KU5.** An overview of Industry 4.0 technologies their use cases and potential applications

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- KU6.** Basic concepts of design thinking agile methodologies and iterative development approaches
- KU7.** Programming constructs coding practices and standards relevant to the chosen programming language
- KU8.** Tools platforms and frameworks necessary for code development testing and deployment in the programming environment
- KU9.** Fundamental techniques for securing software code against vulnerabilities
- KU10.** Tools and methods for documentation data visualization analysis and reporting
- KU11.** Principles of requirements analysis and translating business needs into technical specifications

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Understanding the impact of business related decisions on the organization
- GS2.** Communicating effectively both orally and in writing to convey information clearly
- GS3.** Planning and organizing work efficiently to achieve targets and meet deadlines

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Check the design specifications and define the objectives and functionality of the Java program and align them with design specifications	3	5	-	2
PC2. Develop software code for various applications in Java	3	5	-	2
PC3. Implement coding best practices such as reusability and naming conventions etc.	3	5	-	2
PC4. Create appropriate test cases and test the Java program	3	5	-	2
PC5. Debug the code and review the code with appropriate stakeholders	3	5	-	2
PC6. Maintain a record of test cases and exception handling	3	5	-	2
PC7. Leverage Gen AI tools to automatically generate code snippets for Java programs based on prompts and requirements.	3	5	-	2
PC8. Use Gen AI powered testing tools to create and validate comprehensive test cases for the Java application	3	5	-	2
PC9. Employ Gen AI driven debugging prompts and suggestions to identify bugs and troubleshoot issues efficiently	3	5	-	2
PC10. Utilize Gen AI prompts and suggestions to support automated code reviews and documentation processes	3	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0510
NOS Name	Develop, test and execute software programs as per specifications using Java
Sector	IT-ITeS
Sub-Sector	IT Services
Occupation	Application Deployment
NSQF Level	4
Credits	7
Version	2.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

Qualification Pack

SSC/N0511: Develop, test and execute software programs as per specifications using Python

Description

This NOS is about program development and execution in Python aligned to the prescribed technical specifications

Scope

The scope covers the following :

- Design Verification, Programming and Testing, Documentation

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define the objectives and functionality of the Python program and align it to design specifications
- PC2.** Develop software code for various applications in Python
- PC3.** Implement coding best practices such as reusability and naming conventions etc
- PC4.** Create appropriate test cases and test the Python program
- PC5.** Debug the code and review the code with appropriate stakeholders
- PC6.** Maintain a record of test cases and exception handling etc
- PC7.** Leverage Gen AI tools to automatically generate Python code snippets based on prompts and requirements
- PC8.** Use Gen AI-powered testing tools to create and validate comprehensive test cases for the Python application
- PC9.** Employ Gen AI-driven debugging prompts and suggestions to identify bugs and troubleshoot Python code efficiently
- PC10.** Utilize Gen AI-generated prompts and suggestions to support automated code reviews and generate documentation for Python programs

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational procedures and guidelines related to SDLC programming and documentation
- KU2.** Access and utilize data source documents and information relevant to software development
- KU3.** Identification of stakeholders involved at various stages of the SDLC.
- KU4.** Availability and effective use of standard templates and platforms for coding documentation and project management
- KU5.** An overview of Industry 4.0 technologies their use cases and potential applications

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- KU6.** Basic concepts of design thinking agile methodologies and iterative development approaches
- KU7.** Programming constructs coding practices and standards relevant to the chosen programming language
- KU8.** Tools platforms and frameworks necessary for code development testing and deployment in the programming environment
- KU9.** Fundamental techniques for securing software code against vulnerabilities
- KU10.** Tools and methods for documentation data visualization analysis and reporting
- KU11.** Principles of requirements analysis and translating business needs into technical specifications

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Understanding the impact of business related decisions on the organization
- GS2.** Communicating effectively both orally and in writing to convey information clearly
- GS3.** Planning and organizing work efficiently to achieve targets and meet deadlines

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Define the objectives and functionality of the Python program and align it to design specifications	3	5	-	2
PC2. Develop software code for various applications in Python	3	5	-	2
PC3. Implement coding best practices such as reusability and naming conventions etc	3	5	-	2
PC4. Create appropriate test cases and test the Python program	3	5	-	2
PC5. Debug the code and review the code with appropriate stakeholders	3	5	-	2
PC6. Maintain a record of test cases and exception handling etc	3	5	-	2
PC7. Leverage Gen AI tools to automatically generate Python code snippets based on prompts and requirements	3	5	-	2
PC8. Use Gen AI-powered testing tools to create and validate comprehensive test cases for the Python application	3	5	-	2
PC9. Employ Gen AI-driven debugging prompts and suggestions to identify bugs and troubleshoot Python code efficiently	3	5	-	2
PC10. Utilize Gen AI-generated prompts and suggestions to support automated code reviews and generate documentation for Python programs	3	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0511
NOS Name	Develop, test and execute software programs as per specifications using Python
Sector	IT-ITeS
Sub-Sector	IT Services
Occupation	Application Deployment
NSQF Level	4
Credits	7
Version	2.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

Qualification Pack

SSC/N0512: Write, test and execute programs in HTML5 to develop web or mobile applications as per specifications

Description

This NOS is about code development using HTML5 along with other supporting languages such as JavaScript, CSS, etc.

Scope

The scope covers the following :

- Design Verification, Programming and Testing, Documentation

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define the objectives and functionality of the program and align it to design specifications for the web or mobile app to be developed
- PC2.** Develop and execute software code for various web/ mobile pages and applications using HTML5
- PC3.** Implement coding best practices such as reusability and naming conventions etc.
- PC4.** Create appropriate test cases and test the web or mobile app
- PC5.** Debug the code and review the code with appropriate stakeholders
- PC6.** Maintain a record of test cases and exception handling
- PC7.** Use Gen AI-powered tools to automatically generate HTML5 CSS and JavaScript code snippets based on project requirements and prompts
- PC8.** Employ Gen AI driven testing tools to create and validate test cases for web or mobile applications
- PC9.** Utilize Gen AI powered debugging suggestions to quickly identify and fix issues in HTML CSS and JavaScript code
- PC10.** Leverage Gen AI to generate documentation and comments for HTML5 CSS and JavaScript code to ensure maintainability and clarity

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational procedures and guidelines related to SDLC programming and documentation
- KU2.** Access and utilize data source documents and information relevant to software development
- KU3.** Identification of stakeholders involved at various stages of the SDLC.
- KU4.** Availability and effective use of standard templates and platforms for coding documentation and project management
- KU5.** An overview of Industry 4.0 technologies their use cases and potential applications

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- KU6.** Basic concepts of design thinking agile methodologies and iterative development approaches
- KU7.** Programming constructs coding practices and standards relevant to the chosen programming language
- KU8.** Tools platforms and frameworks necessary for code development testing and deployment in the programming environment
- KU9.** Fundamental techniques for securing software code against vulnerabilities
- KU10.** Tools and methods for documentation data visualization analysis and reporting
- KU11.** Principles of requirements analysis and translating business needs into technical specifications

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Understanding the impact of business related decisions on the organization
- GS2.** Communicating effectively both orally and in writing to convey information clearly
- GS3.** Planning and organizing work efficiently to achieve targets and meet deadlines

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Define the objectives and functionality of the program and align it to design specifications for the web or mobile app to be developed	3	5	-	2
PC2. Develop and execute software code for various web/ mobile pages and applications using HTML5	3	5	-	2
PC3. Implement coding best practices such as reusability and naming conventions etc.	3	5	-	2
PC4. Create appropriate test cases and test the web or mobile app	3	5	-	2
PC5. Debug the code and review the code with appropriate stakeholders	3	5	-	2
PC6. Maintain a record of test cases and exception handling	3	5	-	2
PC7. Use Gen AI-powered tools to automatically generate HTML5 CSS and JavaScript code snippets based on project requirements and prompts	3	5	-	2
PC8. Employ Gen AI driven testing tools to create and validate test cases for web or mobile applications	3	5	-	2
PC9. Utilize Gen AI powered debugging suggestions to quickly identify and fix issues in HTML CSS and JavaScript code	3	5	-	2
PC10. Leverage Gen AI to generate documentation and comments for HTML5 CSS and JavaScript code to ensure maintainability and clarity	3	5	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0512
NOS Name	Write, test and execute programs in HTML5 to develop web or mobile applications as per specifications
Sector	IT-ITeS
Sub-Sector	IT Services
Occupation	Application Deployment
NSQF Level	4
Credits	7
Version	2.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

Qualification Pack

SSC/N0515: Develop, test and execute software programs as per specifications using Kotlin

Description

This NOS is about program development and execution in Kotlin aligned to the prescribed technical specifications

Scope

The scope covers the following :

- Develop, test and execute software programs as per specifications using Kotlin

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand Kotlin syntax and control structures to develop efficient code
- PC2.** Write and execute basic Kotlin programs fulfilling specified requirements with the assistance of Gen AI-generated code snippets
- PC3.** Implement Functions and Modular Code Create reusable functions with appropriate parameters return types and logic
- PC4.** Utilise Kotlin collections and data structures such as lists and arrays for data manipulation supported by Gen AI-driven optimisations
- PC5.** Apply Object-Oriented Programming (OOP) Principles to Design and develop classes and interfaces in Kotlin
- PC6.** Handle Null Safety and Exceptions Implement null safety features and exception handling to ensure robust applications
- PC7.** Use Kotlin development tools effectively including AI-powered features in IDEs for coding and debugging and testing
- PC8.** Develop and execute test cases for Kotlin programs employing AI-driven testing tools to identify bugs and ensure correctness
- PC9.** Adopt Kotlin-specific Features Incorporate features like extension functions and lambdas and coroutines with Gen AI assistance to write efficient code
- PC10.** Generate and maintain comprehensive documentation and comments using Gen AI tools to improve code clarity and maintainability

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational procedures and guidelines related to SDLC programming and documentation
- KU2.** Access and utilize data source documents and information relevant to software development
- KU3.** Identification of stakeholders involved at various stages of the SDLC.

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- KU4.** Availability and effective use of standard templates and platforms for coding documentation and project management
- KU5.** An overview of Industry 4.0 technologies their use cases and potential applications
- KU6.** Basic concepts of design thinking agile methodologies and iterative development approaches
- KU7.** Programming constructs coding practices and standards relevant to the chosen programming language
- KU8.** Tools platforms and frameworks necessary for code development testing and deployment in the programming environment
- KU9.** Fundamental techniques for securing software code against vulnerabilities
- KU10.** Tools and methods for documentation data visualization analysis and reporting
- KU11.** Principles of requirements analysis and translating business needs into technical specifications

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Understanding the impact of business related decisions on the organization
- GS2.** Communicating effectively both orally and in writing to convey information clearly
- GS3.** Planning and organizing work efficiently to achieve targets and meet deadlines

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	20	-
PC1. Understand Kotlin syntax and control structures to develop efficient code	3	5	2	-
PC2. Write and execute basic Kotlin programs fulfilling specified requirements with the assistance of Gen AI-generated code snippets	3	5	2	-
PC3. Implement Functions and Modular Code Create reusable functions with appropriate parameters return types and logic	3	5	2	-
PC4. Utilise Kotlin collections and data structures such as lists and arrays for data manipulation supported by Gen AI-driven optimisations	3	5	2	-
PC5. Apply Object-Oriented Programming (OOP) Principles to Design and develop classes and interfaces in Kotlin	3	5	2	-
PC6. Handle Null Safety and Exceptions Implement null safety features and exception handling to ensure robust applications	3	5	2	-
PC7. Use Kotlin development tools effectively including AI-powered features in IDEs for coding and debugging and testing	3	5	2	-
PC8. Develop and execute test cases for Kotlin programs employing AI-driven testing tools to identify bugs and ensure correctness	3	5	2	-
PC9. Adopt Kotlin-specific Features Incorporate features like extension functions and lambdas and coroutines with Gen AI assistance to write efficient code	3	5	2	-
PC10. Generate and maintain comprehensive documentation and comments using Gen AI tools to improve code clarity and maintainability	3	5	2	-
NOS Total	30	50	20	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0515
NOS Name	Develop, test and execute software programs as per specifications using Kotlin
Sector	IT-ITeS
Sub-Sector	
Occupation	Application Deployment
NSQF Level	4
Credits	7
Version	1.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

Qualification Pack

SSC/N0516: Develop, test and execute software programs as per specifications using Golang

Description

This NOS is about program development and execution in Golang aligned to the prescribed technical specifications

Scope

The scope covers the following :

- Develop, test and execute software programs as per specifications using Golang

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate knowledge of Golang syntax and control structures such as loops and conditionals.
- PC2.** Develop and test and run simple Golang programs that fulfill specified functional requirements including input and output handling and basic logic
- PC3.** Create reusable functions with appropriate parameters and internal logic to promote code modularity and clarity supported by Gen AI based code snippets
- PC4.** Utilize Golang collections and data structures effectively for data storage and manipulation
- PC5.** Apply object oriented principles using structs and interfaces to design scalable and maintainable Go programs
- PC6.** Implement error handling idioms using Golang error type and built-in mechanisms and use goroutines and channels to manage concurrency and ensure thread safety
- PC7.** Use Golang IDEs and development tools proficiently including Gen AI powered features for coding and testing
- PC8.** Develop and validate test cases using Golangs testing package and troubleshoot bugs effectively employing AI-driven debugging suggestions
- PC9.** Incorporate idiomatic Go features such as defer and goroutines supported by Gen AI tools for writing efficient and scalable code
- PC10.** Create programs that align with detailed project specifications ensuring accuracy and maintainability with Gen AI assistance in documentation and optimization

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational procedures and guidelines related to SDLC programming and documentation
- KU2.** Access and utilize data source documents and information relevant to software development
- KU3.** Identification of stakeholders involved at various stages of the SDLC.

Qualification Pack

- KU4.** Availability and effective use of standard templates and platforms for coding documentation and project management
- KU5.** An overview of Industry 4.0 technologies their use cases and potential applications
- KU6.** Basic concepts of design thinking agile methodologies and iterative development approaches
- KU7.** Programming constructs coding practices and standards relevant to the chosen programming language
- KU8.** Tools platforms and frameworks necessary for code development testing and deployment in the programming environment
- KU9.** Fundamental techniques for securing software code against vulnerabilities
- KU10.** Tools and methods for documentation data visualization analysis and reporting
- KU11.** Principles of requirements analysis and translating business needs into technical specifications

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Understanding the impact of business related decisions on the organization
- GS2.** Communicating effectively both orally and in writing to convey information clearly
- GS3.** Planning and organizing work efficiently to achieve targets and meet deadlines

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Demonstrate knowledge of Golang syntax and control structures such as loops and conditionals.	3	5	-	2
PC2. Develop and test and run simple Golang programs that fulfill specified functional requirements including input and output handling and basic logic	3	5	-	2
PC3. Create reusable functions with appropriate parameters and internal logic to promote code modularity and clarity supported by Gen AI based code snippets	3	5	-	2
PC4. Utilize Golang collections and data structures effectively for data storage and manipulation	3	5	-	2
PC5. Apply object oriented principles using structs and interfaces to design scalable and maintainable Go programs	3	5	-	2
PC6. Implement error handling idioms using Golang error type and built-in mechanisms and use goroutines and channels to manage concurrency and ensure thread safety	3	5	-	2
PC7. Use Golang IDEs and development tools proficiently including Gen AI powered features for coding and testing	3	5	-	2
PC8. Develop and validate test cases using Golangs testing package and troubleshoot bugs effectively employing AI-driven debugging suggestions	3	5	-	2
PC9. Incorporate idiomatic Go features such as defer and goroutines supported by Gen AI tools for writing efficient and scalable code	3	5	-	2
PC10. Create programs that align with detailed project specifications ensuring accuracy and maintainability with Gen AI assistance in documentation and optimization	3	5	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0516
NOS Name	Develop, test and execute software programs as per specifications using Golang
Sector	IT-ITeS
Sub-Sector	
Occupation	Application Deployment
NSQF Level	4
Credits	7
Version	1.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N0509.Verify the specifications and contribute to the design of software program	30	50	-	20	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	15
Total	50	80	-	20	150	30

Elective: 1 Java

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N0510.Develop, test and execute software programs as per specifications using Java	30	50	-	20	100	70
Total	30	50	-	20	100	70

Elective: 2 Python

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N0511.Develop, test and execute software programs as per specifications using Python	30	50	-	20	100	70
Total	30	50	-	20	100	70

Elective: 3 HTML 5

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N0512. Write, test and execute programs in HTML5 to develop web or mobile applications as per specifications	30	50	-	20	100	70
Total	30	50	-	20	100	70

Elective: 4 Kotlin

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N0515. Develop, test and execute software programs as per specifications using Kotlin	30	50	20	-	100	70
Total	30	50	20	-	100	70

Elective: 5 Golang

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N0516. Develop, test and execute software programs as per specifications using Golang	30	50	-	20	100	70
Total	30	50	-	20	100	70

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.